

CYBERSECURITY SUITE (3-IN-1)

ADVANCED SYSTEM MONITORING & SECURITY TECHNICAL DOCUMENTATION

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1. EXECUTIVE SUMMARY

The **CyberSecurity Suite (3-in-1)** is an integrated desktop security application built using Python. It incorporates three major defensive and operational security modules: a **Password Strength Analyzer** with breach verification, a real-time **Keylogger Activity Monitor** with remote notification pipelines (Email & Telegram), and a robust **Text Encryption Suite** utilizing AES cryptography.

2. SYSTEM ARCHITECTURE & CODEBASE ENVIRONMENT

The software is structured with high modularity in Visual Studio Code, utilizing object-oriented Python design patterns. Multi-threading is employed to isolate UI execution from background logging processes and network transmissions.

VS Code IDE Architecture

Figure 1: Visual Studio Code IDE workspace showing main execution logic (main.py) and module structure.

3. FUNCTIONAL MODULES & SCREENSHOT VERIFICATIONS

Module A: Password Strength Analyzer

Analyzes entered credentials for length, entropy, and pattern complexity. It computes SHA-1 hashes and executes real-time k-Anonymity API queries against the *HaveIBeenPwned* database to detect compromised credentials safely.

Password Analyzer Screen

Figure 2: Password Strength Analyzer detecting an unsafe, breached password (Score 0/5).

Module B: Keylogger & Remote Activity Monitor

Tracks user input events in real time. It features integrated configuration panels for SMTP email transmission and Telegram Bot API automated alerting (logs and full desktop screenshots).

Telegram Bot Handshake Test

Figure 3: Telegram Bot connectivity test confirming active API handshake.

Keylogger Execution Monitor

Figure 4: Live Keylogger monitor running and logging keystrokes dynamically.

Module C: Remote Telegram Integration & Automated Screenshot Delivery

The keylogger automatically formats captured key events and transmits structured reports via Telegram. Upon triggering specific execution signals or stop commands, an automated system screenshot is captured and sent directly to the designated Telegram chat.

Telegram Keystroke Log 1

Figure 5: Keystroke batch log output.

Telegram Keystroke Log 2

Figure 6: Keystroke & backspace audit.

Telegram Auto-Screenshot Delivery

Figure 7: Automated desktop screenshot capture.

Module D: Text Encryption Suite

Provides end-to-end data confidentiality by encrypting sensitive plaintext into AES ciphertext (Base64 encoded) and supporting instant decryption using secret keys.

Text Encryption Module

Figure 8: Text Encryption Suite performing AES plaintext-to-ciphertext conversion.

4. TECHNICAL SPECIFICATIONS

Module / Subsystem	Technology Stack	Core Responsibility
User Interface	Python Tkinter / ttk.Notebook	Responsive desktop tabbed navigation.
Credential Security	SHA-1 / HIBP REST API	k-Anonymity privacy-preserving breach checks.
Input Monitoring	Pynput & Threading	Asynchronous global keystroke capture.
Data Cryptography	PyCryptodome (AES-GCM / CBC)	Strong symmetric data encryption and decryption.
Remote Telemetry	Requests & Telegram Bot API	Automated event logs and screenshot dispatching.

5. CONCLUSION

The **CyberSecurity Suite** successfully integrates credential assessment, cryptographic data protection, and real-time remote system telemetry into a single, cohesive Python desktop utility.